

# “Bird” Over Time: a Time Series Analysis of Charlie Parker’s Harmonic Complexity

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## 1. Abstract

This study presents a computational time series analysis of Charlie Parker’s improvisational practice using the complete Charlie Parker Omnibook corpus. We develop metrics for harmonic complexity, dissonance, and rate of change, analyzing temporal patterns through phrase-level segmentation using actual rest boundaries. We identify three distinct improvisational strategies and introduce theoretical concepts including *gravity* (causal relationships via Granger testing) and *robustness* (triadic content).

Our analysis reveals fundamental temporal heterogeneity: 80.4% of performances show no phrase-level causality between dissonance and complexity, suggesting chorus-level architectural planning predominates over reactive navigation. Strategy-specific patterns demonstrate distinct organizational modes: “Exploratory” performances exhibit near-zero causality (3.6%), consistent with composed improvisation; “Balanced” performances show moderate reactive effects (11.0%); “Contrasting” performances emphasize phrase-length modulation (23.1%). Among tunes exhibiting causality, dissonance predicts subsequent complexity more frequently than the reverse (17.4% vs 8.7%), though this asymmetry is suggestive rather than definitive ( $p = 0.109$ ). Parker’s vocabulary shows significant triadic sparsity correlated with frequency ( $r = -0.350$ ,  $p < 0.001$ , Cohen’s  $d = -0.87$ ), suggesting mastery through navigational flexibility via triadically ambiguous structures rather than exotic material accumulation.

**Keywords:** Charlie Parker, computational musicology, time series analysis, interval vectors, jazz improvisation, Granger causality

## 2. Introduction

The temporal evolution of harmonic material during jazz improvisation remains incompletely understood despite extensive theoretical and pedagogical attention. Thomas Owens’s seminal dissertation (Owens 1974) established comprehensive analysis of Parker’s melodic vocabulary through pitch-pattern cataloging, revealing Parker’s reliance on a finite set of formulaic patterns deployed across multiple harmonic contexts. While significant research addresses what materials improvisers use, examining solos as static collections of pitch materials or chord-scale relationships, far less examines how those materials unfold temporally. Existing theoretical frameworks focus on vocabulary and pattern storage but leave underspecified the temporal dynamics of deployment during performance. Charlie Parker’s improvisational language, in particular, exhibits sophisticated temporal organization that has remained resistant to quantitative analysis.

Recent work applying network science to jazz (Merseal et al. 2022) demonstrates that melodic sequences can be represented as networks where nodes represent pitch patterns and edges represent temporal continuations. This provides a spatial metaphor (network topology) for understanding temporal sequences. However, network approaches emphasize static structural properties rather than temporal evolution within individual performances. Our time series approach complements both pattern-based analysis and network analysis by foregrounding temporal dynamics: how does complexity evolve within a solo? Do improvisers build toward climaxes or maintain consistency? Are there causal relationships where harmonic conditions at moment  $t$  predict choices at moment  $t + 1$ ? These questions require time series methodology.

This study employs time series analysis to examine Parker’s complete recorded corpus as represented in the Charlie Parker Omnibook, treating improvisation as a temporal process where harmonic states evolve according to specific patterns and strategies. Rather than analyzing static harmonic vocabulary, we track how interval vectors (i.e. mathematical representations of harmonic content) change from chord to chord, revealing temporal patterns invisible to traditional analysis.

A methodological note on terminology: while Parker’s saxophone produces monophonic melodic lines, our analysis examines the harmonic implications of those melodic choices. We aggregate pitch classes into interval vectors: mathematical representations of intervallic relationships which capture the harmonic content implied by melodic improvisation at different temporal scales. Thus the term “harmonic” refers to pitch-class density, not to simultaneously sounded notes. Through this methodology, we identify three distinct improvisational strategies ranging from balanced continuity to episodic contrast, revealing that Parker conceives architectural approaches per composition rather than employing a uniform improvisational method.

Our approach introduces metrics for quantifying harmonic evolution across temporal scales, enabling statistical analyses that reveal causal relationships between harmonic dimensions, what we term *gravity* (measured via Granger causality testing), and structural flexibility through triadic content, which we term *robustness*. The analysis proceeds through multiple temporal scales:

- segment-level (individual chord changes)
- phrase-level (detected using actual rest boundaries from scores)
- half-chorus level (rolling windows)
- full-chorus level (comparing temporal evolution across choruses)

This multi-scale approach reveals organizational principles operating at different temporal granularities, from rapid phrase articulation to chorus-level solo development arcs.

### 3. Methodology

#### 3.1. Time series construction and analysis

We analyze the Charlie Parker Aligned Digital Omnibook, comprising 50 transcribed performances (composed melodies and improvisations) with aligned MusicXML and MIDI data. The dataset combines manual transcriptions created by Déguernel, Vincent, and Assayag (Déguernel et al. 2016) with automatic transcription methods developed by Riley and Dixon (Riley and Dixon 2024). This corpus provides:

- Complete pitch and rhythm transcriptions
- Aligned MIDI timestamps enabling temporal analysis
- Chord progression annotations
- Chorus boundary markings
- Form structure annotations (AABA, 12-bar blues, etc.)

The aligned nature of this dataset is crucial: MusicXML provides accurate pitch and rest information, while MIDI alignment provides precise timestamps, enabling true time series analysis rather than relying on beat position alone.

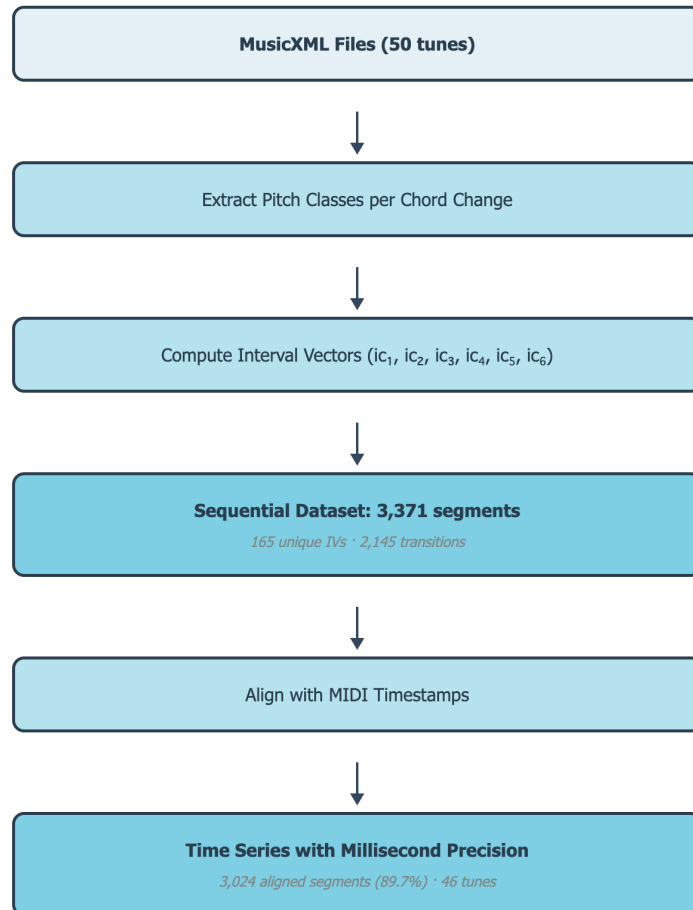


Figure 1: computational pipeline for time series construction over the Omnibook corpus

The pipeline processed all 50 tunes successfully, extracting 3,371 harmonic segments. We identified 165 unique interval vectors and 2,145 unique transitions. MIDI alignment successfully mapped 3,024 segments (89.7%) to precise timestamps; 4 tunes failed alignment due to missing MIDI files (“Anthropology”, “Shaw ‘Nuff“, “Segment”, and “Chasing the Bird”) but retained all other analytical data. Subsequent analyses requiring precise timestamps (rolling windows, autocorrelation, Granger causality) were performed on the 46 successfully aligned tunes (3,024 segments).

The four excluded tunes represent significant compositions in Parker’s repertoire, particularly “Anthropology” and “Chasing the Bird”, both archetypal bebop contrafacts. While their exclusion from temporal analyses limits generalization, vocabulary and transition analyses include these tunes, ensuring core findings about Parker’s harmonic vocabulary remain comprehensive.

For each chord change in the corpus, we extract all pitch classes sounding over that harmonic moment and compute the interval vector. The interval vector is a six-dimensional vector ( $ic_1, ic_2, ic_3, ic_4, ic_5, ic_6$ ) where each component counts occurrences of each interval class. This abstracts away from register and specific voicing while preserving harmonic content. Successive interval vectors, aligned with their MIDI timestamps, create a time series representation of harmonic evolution. This enables standard time series analysis techniques while maintaining musical relevance: interval vectors capture harmonic information essential to jazz theory.

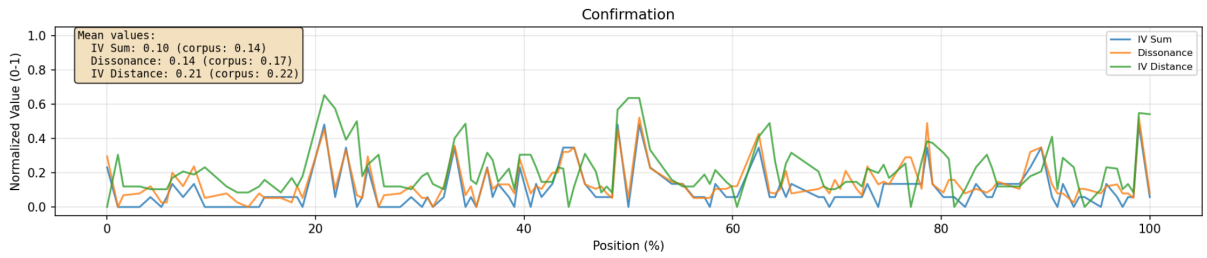


Figure 2: temporal evolution of harmonic metrics for “Confirmation.” IV sum (complexity), dissonance, and IV distance (rate of change) are normalized to [0,1] scale. Position normalized to percentage enables comparison across performances of different lengths. Green spikes indicate rapid harmonic shifts; blue and orange show concurrent changes in complexity and dissonance.

Mean values shown relative to corpus averages.

We computed three main metrics for each temporal segment:

- **Complexity (iv\_sum):**  $\sum_{i=1}^6 ic_i$

Total interval content reflecting harmonic density. Higher values indicate more harmonic content, generally corresponding to more complex pitch-class sets.

- **Dissonance:**  $ic_1 + 0.5 \cdot ic_2 + 0.8 \cdot ic_6$

Weighted sum emphasizing dissonant intervals: minor seconds ( $\times 1.0$ ), tritones ( $\times 0.8$ ), and major seconds ( $\times 0.5$ ). Consonant intervals (thirds and fourths) are excluded as they do not contribute to perceptual roughness.

These weights reflect psychoacoustic principles of sensory dissonance as measured by critical band interference (Plomp and Levelt 1965; Sethares 1993). Minor seconds ( $ic_1$ ) receive maximum weight (1.0) due to maximal critical band overlap. Tritones ( $ic_6$ ) receive 0.8 weight reflecting their tonal instability and historical treatment as requiring resolution,

though with less perceptual roughness than minor seconds. Major seconds (ic2) receive 0.5 weight as they produce moderate critical band interference. Perfect fourths and fifths (ic5) and major and minor thirds (ic3, ic4) receive zero weight as these intervals fall outside critical band overlap and function as consonances in tonal and post-tonal contexts. This weighting scheme prioritizes sensory dissonance (roughness) over tonal dissonance (functional tension), appropriate for analyzing melodic improvisation where vertical harmonic function is implied rather than explicit.

- **Intervallic distance (iv\_distance):**  $\sqrt{\sum_{i=1}^6 (ic_i^{t+1} - ic_i^t)^2}$

It's the Euclidean distance between consecutive interval vectors. It measures rate of harmonic change: how much harmonic content transforms between consecutive moments.

These metrics reduce six-dimensional interval vectors to interpretable scalar values while preserving essential harmonic information. They also enable standard statistical analysis (mean, standard deviation, correlation) while remaining musically meaningful.

## 3.2. Multi-scale pattern analysis

### 3.2.1. Phrase Detection

Having established segment-level time series, we next identified phrases using actual rests from musical scores rather than inferring pauses from timestamp gaps. We parsed MusicXML `<rest>` elements with their precise durations and positions, matching them to temporal segments.

This approach respects Parker's actual articulation as notated by expert transcribers. Bebop phrasing is highly articulated with frequent short rests; inferring phrases from timestamp gaps would miss most phrase boundaries.



Figure 3: Phrase detection in “Si Si”

Rests meeting a minimum threshold of 0.5 quarter notes (eighth note or longer) were designated as phrase boundaries. This threshold was selected based on Parker's articulation style: bebop phrasing employs frequent short rests as phrase punctuation rather than breath necessity. Pilot

testing with alternative thresholds (0.25, 0.5, and 1.0 quarter notes) revealed that 0.25 captured excessive micro-articulations within continuous melodic lines, while 1.0 missed legitimate phrase boundaries in faster tempos. The 0.5 quarter note threshold balanced sensitivity (capturing meaningful phrase articulation) with specificity (avoiding within-phrase micro-pauses). This threshold yielded 1,507 phrases across 46 tunes (mean = 32.8 phrases per tune), consistent with typical bebop phrase density where solo choruses contain 8-12 phrases per 32-bar form.

For each phrase, we aggregated segment-level metrics into phrase-level statistics: *mean complexity* (mean\_iv\_sum), mean dissonance, *phrase length* in segments (segment\_count), and standard deviations. This yielded 1,507 phrases across 46 tunes. Phrases serve as the fundamental analytical unit, representing musically meaningful temporal segments while avoiding the extremes of overly granular (segment-level) or coarse (chorus-level) analysis.

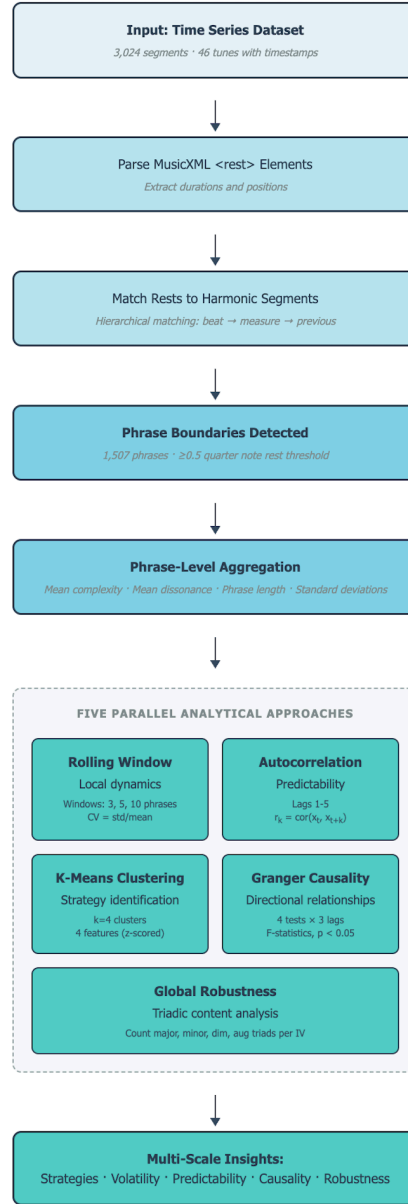


Figure 4: after phrase detection, five parallel approaches examine temporal patterns at different scales: local dynamics (rolling windows), predictability (autocorrelation), strategies (clustering), causality (Granger testing), and structural properties (robustness)

### 3.2.2. Rolling window analysis

To capture local temporal dynamics, we computed *rolling window statistics* over phrase sequences using window sizes of 3, 5, and 10 phrases. For each window position, we calculated rolling mean, rolling standard deviation, and coefficient of variation ( $CV = \text{std}/\text{mean}$ ). The CV quantifies local volatility: high CV indicates rapid fluctuations in complexity or dissonance within a small temporal window, while low CV suggests stable regions.

Window size selection reflects different temporal scales:

- 3-phrase windows capture immediate phrase-to-phrase volatility
- 5-phrase windows provide balanced smoothing (recommended for most analyses)
- 10-phrase windows reveal macro-level trends approaching half-chorus scale.

Rolling statistics were centered (computed symmetrically around each phrase) to avoid introducing artificial temporal lag.

### 3.2.3. Autocorrelation and predictability

We also computed *autocorrelation* to quantify phrase-to-phrase predictability. For each tune, *complexity* and *dissonance* time series were analyzed using Pearson correlation between the series and lagged versions of itself (lags 1 through 5). Positive autocorrelation indicates continuous development (high phrase predicts high next phrase), negative autocorrelation indicates contrasting structure (high phrase followed by low phrase), and near-zero autocorrelation suggests random walk or episodic organization.

For a time series  $\{x_t\}_{t=1}^n$  and lag  $k$ :

$$r_k = \text{cor}(x_t, x_{t+k}) \quad \text{for } t = 1, \dots, n - k$$

Tunes with fewer than 6 phrases were excluded to ensure sufficient statistical power.

### 3.2.4. Improvisational strategy clustering

To identify distinct improvisational strategies, we performed *K-means clustering* using four features per tune:

- mean complexity (mean\_iv\_sum\_mean) - average harmonic density
- complexity volatility (mean\_iv\_sum\_cv) - coefficient of variation across phrases
- predictability (mean\_iv\_sum\_lag1) - autocorrelation at lag-1
- dissonance volatility (mean\_dissonance\_cv) - dissonance variation

Features were standardized (z-scored) before clustering to ensure equal weighting. We selected  $k=4$  clusters based on interpretability and separation in PCA space. The choice of  $k=4$  clusters balances statistical separation with interpretability. Silhouette analysis suggested  $k=3-5$  as plausible values (silhouette scores:  $k=3$ : 0.42,  $k=4$ : 0.38,  $k=5$ : 0.35), but  $k=4$  provided the most musically interpretable clusters with distinct temporal profiles. We prioritize interpretability over optimal separation, as our goal is identifying meaningful improvisational strategies rather than maximizing classification accuracy. K-means was initialized with 20 random starts to avoid local optima.

The resulting clusters represent distinct approaches to temporal organization: “Balanced” (Cluster 0, 25 tunes), “Contrasting” (Cluster 1, 13 tunes), “Exploratory” (Cluster 2, 7 tunes), and one outlier (Cluster 3, 1 tune).

### 3.2.5. Granger causality analysis

We tested directional *causal relationships* between harmonic dimensions using Granger causality. Four hypotheses were examined:

- Complexity  $\rightarrow$  Dissonance (does complexity create tension?)
- Dissonance  $\rightarrow$  Complexity (does dissonance trigger complexity?)
- Phrase Length  $\rightarrow$  Complexity (does longer phrase enable complexity?)
- Phrase Length  $\rightarrow$  Dissonance (does longer phrase enable dissonance?)

For each tune with  $\geq 10$  phrases, we performed Granger causality tests using the statsmodels implementation of the F-test for model comparison. The null hypothesis states that past values of the cause variable do not improve prediction of the effect variable beyond its own autoregressive history.

Tests were conducted at lags 1, 2, and 3 phrases. Lag-1 captures immediate causal effects (phrase  $t$  predicts phrase  $t + 1$ ), while lags 2-3 capture delayed effects. F-statistics quantify causal strength: higher values indicate stronger predictive relationships.

Results are reported as: (1) percentage of tunes showing significant causality per test, (2) mean F-statistic across corpus, and (3) cluster-specific patterns revealing strategy-dependent causal structures.

### 3.2.6. Robustness

We also explored *robustness* as a metric that quantifies triadic content: the number of major, minor, diminished, and augmented triads contained within each interval vector. This metric captures structural flexibility: IVs with higher triadic content offer more options for upper structure reharmonization and voice-leading transformations.

For each unique interval vector in the corpus, we performed exact enumeration of all pitch-class sets generating that interval vector at the estimated cardinality (derived from IV sum via  $n(n-1)/2 = \Sigma IC$ ). For each candidate PC set, we enumerated all 3-note subsets and identified those matching standard triad patterns: major  $[0,4,7]$ , minor  $[0,3,7]$ , diminished  $[0,3,6]$ , and augmented  $[0,4,8]$ . Mean triadic content per IV was computed by averaging triad counts across all generating PC sets. This approach provides exact counts for all cardinalities ( $n = 165$  unique IVs), with results cached for computational efficiency.

We then analyzed the relationship between triadic content and usage frequency across the corpus. For each IV, we recorded its total frequency of occurrence across all performances and computed both unweighted (mean across all IVs) and frequency-weighted (mean weighted by actual usage) triadic content. This reveals whether Parker systematically favors triadically dense or sparse structures in his vocabulary selection.

## 4. Results

### 4.1. Vocabulary

Parker’s improvisational vocabulary centers on a relatively compact set of interval vectors with distinctive functional distributions. At this stage of analysis, each interval vector represents the pitch-class content aggregated over a single chord segment, one IV per chord change. The interval vector (111000), containing one instance each of interval classes 1, 2, and 3 (minor second, major second, minor third), appears 226 times across the corpus, making it the single most frequent harmonic structure (Table 1). This three-note chromatic cluster (Forte class 3-2) can manifest as tightly packed chromatic voicings like {C, C#, E♭} or {C, D, E♭}, or as wider-spaced inversions like {C, A, B♭}. It appears predominantly in II contexts (51 occurrences), though it distributes across multiple functions including tonic (I: 48) and dominant (V: 44), revealing its functional versatility as a pivot structure rather than a function-specific sonority. Over a Dm7 chord, these voicings can function as altered upper extensions (b7, M7, b9), chromatic voice leading (b7, root, b9), or diatonic extensions (b7, 5, 6), explaining the IV’s adaptability across harmonic contexts.

The top ten interval vectors account for 1,155 occurrences (34% of the corpus), suggesting concentrated reliance on a core harmonic vocabulary. Notably, nine of the ten most frequent IVs show primary tonic (I) function, suggesting that Parker’s melodic complexity concentrates at moments of harmonic arrival rather than during transitions. The second most frequent IV, (122010), appears 163 times primarily in tonic contexts (I: 45, V: 37). This four-note tetrachord (Forte class 4-27) typically manifests as shell voicings like {C, D, E♭, B♭} that omit the third, creating harmonic stability without committing to major or minor quality. This ambiguity enables (122010) to function effectively as both major seventh (root, 9, 9, b7) and minor seventh (b7, root, b9, 6) sonorities, providing the inverse distribution of (111000): stable where (111000) is transitional.

Transitional patterns reinforce this vocabulary structure (Table 2). The most common transition, (111000) → (122010) (II → I, 19 occurrences), represents Parker navigating from chromatic cluster to shell voicing, paralleling the archetypal II-V-I resolution by moving from chromatic tension to ambiguous stability. However, the corpus also shows substantial same-IV persistence: (111000) → (111000) appears 18 times, suggesting Parker’s willingness to sustain chromatic clusters across multiple chord changes rather than treating them as fleeting passing structures. Reverse motion also appears: (122010) → (111000) (I → II, 17 occurrences) shows Parker generating harmonic tension by moving from stable shell voicings to altered chromatic clusters, contradicting traditional voice-leading pedagogy that emphasizes tension→resolution directionality.

The bidirectional nature of these transitions, both II → I (19) and I → II (17) ranking in the top three, reveals a fundamental finding: Parker conceives these structures not as functional progressions but as **navigational options** within a network space. The interval vector (111000) functions as a hub not because of its harmonic stability but because of its **positional centrality**, offering maximum continuation options to diverse harmonic destinations. This observation motivates our subsequent time series analysis, which examines how Parker deploys this core vocabulary across temporal scales from phrase-level articulation to full-chorus development arcs.

## 4.2. Top 10 most frequent IVs

| IV       | Frequency | Primary Function | Function Distribution                         |
|----------|-----------|------------------|-----------------------------------------------|
| (111000) | 226       | II               | II: 51, I: 48, V: 44, IV: 24, VI: 19, III: 19 |
| (122010) | 163       | I                | I: 45, V: 37, II: 24, IV: 18, III: 14         |
| (001110) | 121       | I                | I: 48, II: 23, V: 18, III: 9, VI: 8, IV: 8    |
| (122230) | 116       | I                | I: 38, II: 29, IV: 15, V: 13, bIII: 10        |
| (021120) | 98        | I                | I: 24, IV: 22, V: 19, II: 17, bVII: 6         |
| (011010) | 94        | I                | I: 39, II: 19, IV: 11, V: 11, bVII: 6         |
| (121110) | 92        | I                | I: 24, V: 22, II: 12, IV: 11, bIII: 8         |
| (111120) | 90        | I                | I: 30, II: 18, VI: 13, IV: 9, V: 9            |
| (012120) | 81        | I                | I: 21, II: 18, IV: 13, V: 8, III: 5, bIII: 5  |
| (132130) | 74        | I                | I: 21, IV: 17, II: 14, bIII: 7, V: 5          |

Table 1: most frequent interval vectors in Parker corpus with harmonic function distributions

## 4.3. Top 10 most frequent transitions

| Source IV | Function |   | Target IV | Function | Count |
|-----------|----------|---|-----------|----------|-------|
| (111000)  | II       | → | (122010)  | I        | 19    |
| (111000)  | II       | → | (111000)  | II       | 18    |
| (122010)  | I        | → | (111000)  | II       | 17    |
| (001110)  | I        | → | (111111)  | V        | 17    |
| (122010)  | I        | → | (122010)  | I        | 13    |
| (111000)  | II       | → | (122230)  | I        | 11    |
| (111120)  | I        | → | (111000)  | II       | 9     |
| (121110)  | I        | → | (111000)  | II       | 9     |
| (122230)  | I        | → | (111000)  | II       | 9     |
| (112101)  | VI       | → | (101220)  | II       | 8     |

Table 2: most frequent interval vector transitions showing predominant harmonic progressions

#### 4.4. Complexity distribution between segments

Analysis of 3,371 segments across 50 tunes reveals wide variation in harmonic complexity. Mean *complexity* (*iv\_sum*) is 10.01 (SD = 7.18), suggesting substantial variation both within and between tunes.

- **Tunes with most complexity:**

- “Cosmic Rays”: 16.28
- “Bluebird”: 15.85
- “Laird Baird”: 13.86

- **Tunes with least complexity:**

- “Si Si”: 7.09
- “Moose the Mooche”: 7.21
- “Yardbird Suite”: 7.57

Importantly, complexity and variability correlate strongly ( $r = 0.68$ ): complex tunes tend to show more variation in complexity across their temporal evolution. However, this relationship is not deterministic: some tunes break this pattern, as discussed below.

We also explored other metrics such as *dissonance* (mean = 3.2) and *rate of change* (mean = 3.95). The first shows a heavily right-skewed distribution: most segments are relatively consonant (modal dissonance around 1-2), with a long tail extending to highly dissonant passages (up to 18). This suggests Parker’s harmonic language centers on moderate dissonance levels, deploying extreme dissonance strategically rather than consistently. *Rate of change* (*iv\_distance*) shows moderate harmonic change is most common, with occasional shifts. The mean of 4 indicates that typical chord-to-chord motion involves changing approximately 4 units across the six-dimensional interval vector space, neither static repetition nor extreme transformation.

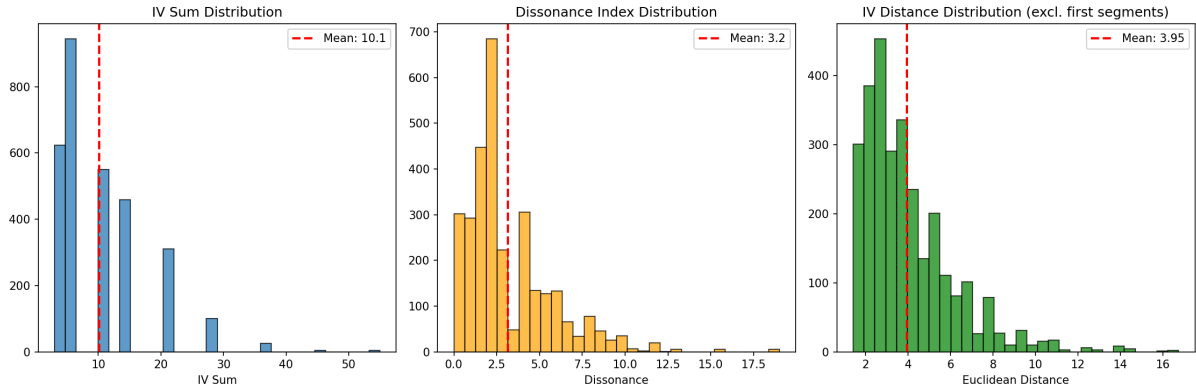


Figure 5: corpus-wide complexity distribution; please note that the first segment of each tune is excluded from rate-of-change analyses as it has no prior segment for comparison.“

## 4.5. Complexity distribution between choruses

Cross-chorus analysis revealed the following patterns:

- Chorus 1 (head) mean complexity: 8.32
- Chorus 2 mean complexity: 11.31
- Subsequent choruses: maintain 10-11

Composed melodies (heads) are often less complex than improvisation choruses. Parker’s first improvised chorus (Chorus 2) already exhibits peak complexity, showing no gradual build within improvisation. Heads contain most vocabulary, and improvisation adds minimal new harmonic content but explores new pathways.

This pattern requires careful interpretation: Parker’s composed melodies (heads) are themselves “composed solos” or “contrafacts” over standard progressions (for example “Anthropology” over “I Got Rhythm” by G. Gershwin). The lower *complexity* likely reflects melodic construction prioritizing memorable contour and singability over maximal harmonic density. Heads establish the core vocabulary while maintaining thematic clarity; improvised choruses then explore denser harmonic regions enabled by the absence of melodic memorability constraints.

Parker’s core improvisational vocabulary is largely established in Chorus 1 (heads). Chorus 2 contains 81.1% of all IVs used across the entire performance, with only 16.5% added during Chorus 1 and minimal additions thereafter (Figure 6). Temporal evolution thus reflects primarily the exploration of new pathways connecting existing harmonic nodes rather than progressive vocabulary expansion.

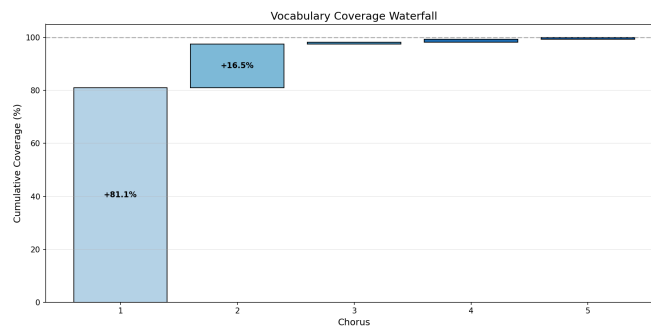


Figure 6: cumulative vocabulary coverage across choruses. Chorus 1 (head) contains 81.1% of all unique interval vectors used in the performance, with only 16.5% added during Chorus 2 (first improvised chorus) and minimal additions thereafter.

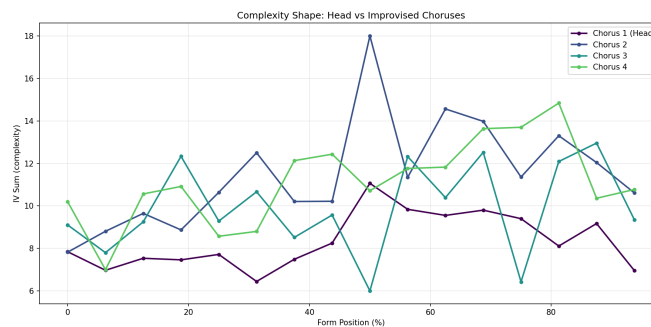


Figure 7: cross-chorus complexity comparison showing head is simpler than improvised choruses

## 4.6. Complexity distribution between tunes

We used Dynamic Time Warping to compare temporal evolution patterns between tunes. DTW allows comparison of sequences of different lengths by finding optimal alignment, analogous to aligning two melodies that occur at different tempos.

Comparing “Cosmic Rays” and “Bluebird” (both high-complexity tunes) reveals dramatically different temporal shapes:

**“Cosmic Rays”:** Episodic structure with multiple peaks and valleys throughout. High volatility but no single climax.

**“Bluebird”:** “Slow-burn climax” structure. Stays relatively simple for first 60%, then explodes at 70-90% with sustained high complexity.

DTW distance between these tunes is large despite similar mean complexity, revealing that Parker doesn’t use a fixed template. Some performances are journeys (building to climax), others are episodes (multiple peaks), others maintain consistency.

This finding suggests Parker conceives each tune’s improvisational arc holistically rather than applying a single approach across all contexts.

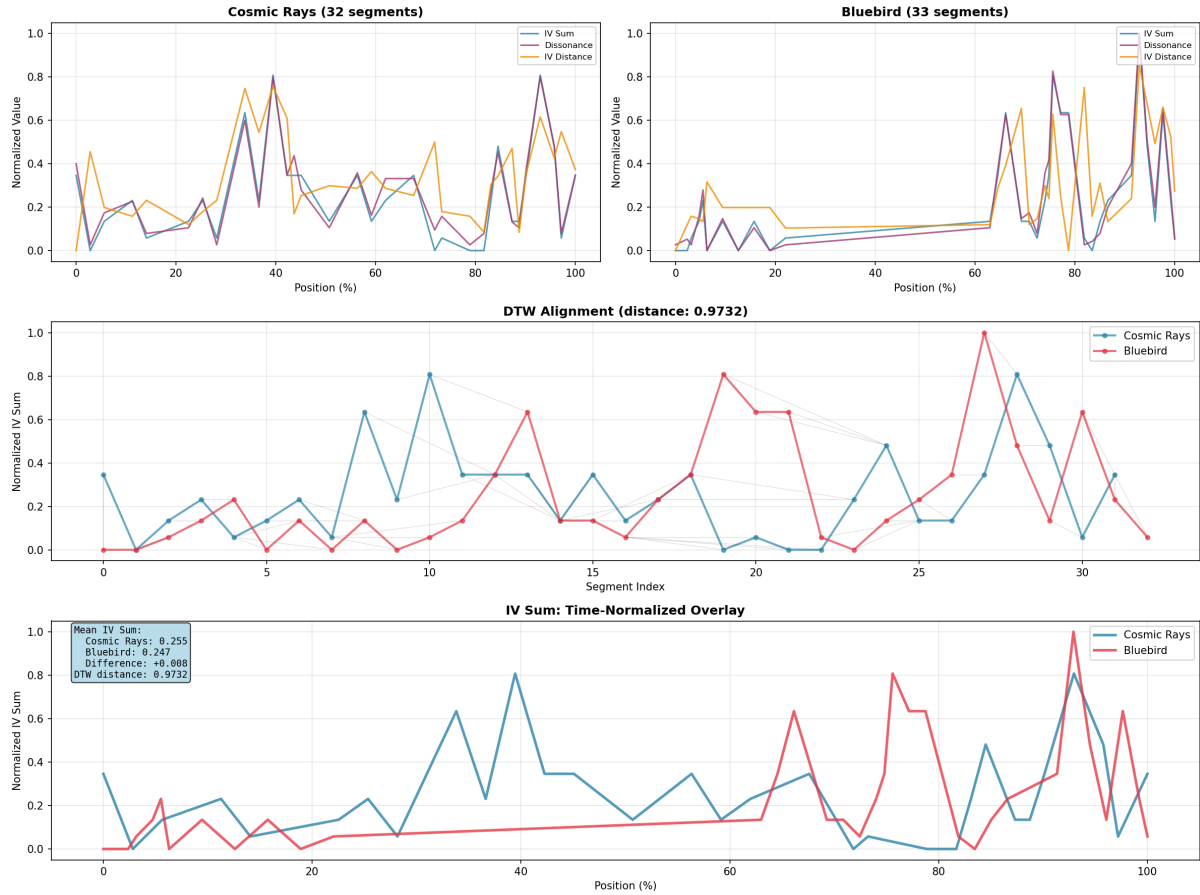


Figure 8: comparing composed melody and solo development between “Cosmic Rays” and “Bluebird”

## 4.7. Complexity distribution between phrases

Rolling window analysis reveals meaningful patterns emerge at 5-phrase window level (usually half-chorus). Surface articulation organizes into phrase groups that constitute functional units of harmonic motion. This suggests hierarchical organization: rapid surface articulation → phrase groups → chorus-level structure.

One tune exhibits paradoxical temporal properties:

### My Little Suede Shoes

- Lowest mean complexity in corpus: 6.58
- Highest complexity variability:  $CV = 0.99$
- Highest dissonance variability:  $CV = 1.37$

This tune alternates between extremely simple and moderately complex passages, exhibiting what we might call a “trap door” structure. Unlike other tunes where complexity relates to volatility, “My Little Suede Shoes” achieves high volatility through contrasts rather than sustained complexity.

This finding demonstrates that temporal organization strategies are tune-specific, not performer-general. Parker conceives different architectural approaches for different compositions.

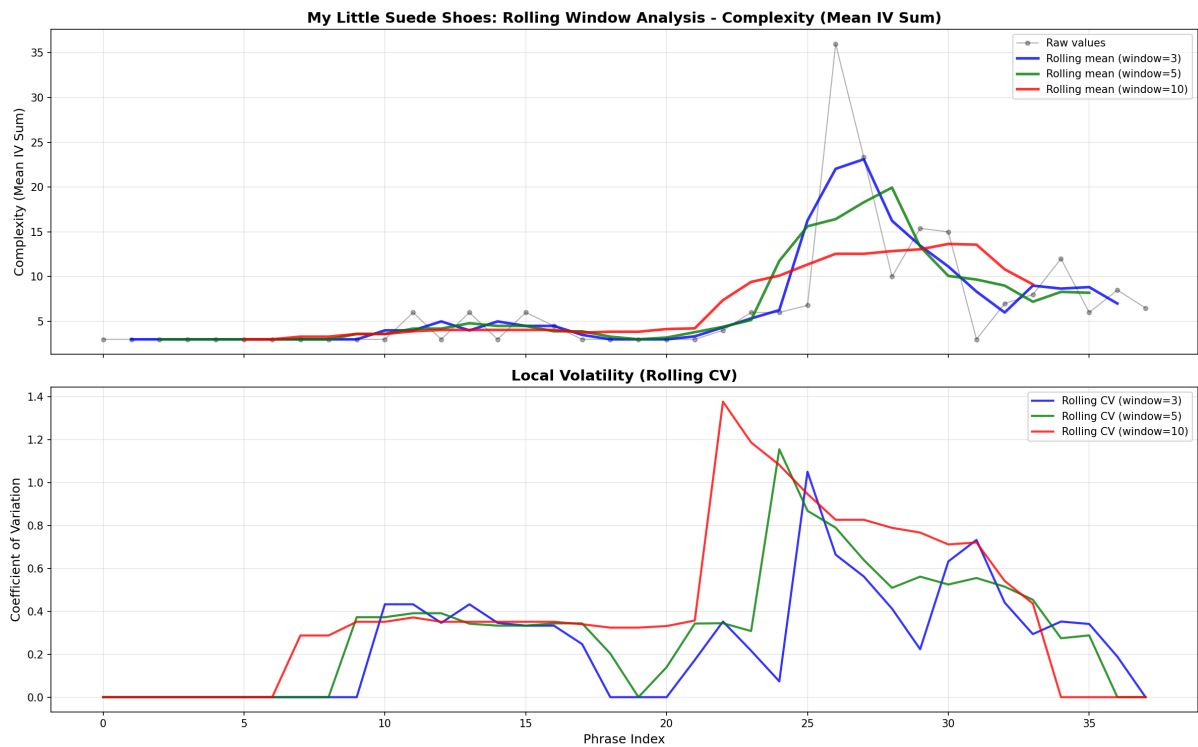


Figure 9: the outlier “My Little Suede Shoes” showing extreme alternation between simple and complex passages“

## 4.8. Complexity distribution over form

We also analyzed whether complexity systematically varies depending on the form of a composition. For AABA tunes, does the bridge (B section) show higher complexity? For blues, does complexity peak at turnaround?

Results show significant formal patterns:

- Second half of forms is more complex than first half (10.87 vs 9.49,  $p < 0.0001$ )
- Bridge sections (bars 17-24 in AABA) peak at 11.27
- Overall pattern: BUILD-SUSTAIN rather than build-decay

Cross-chorus analysis shows Parker maintains high complexity throughout improvisation rather than building gradually across choruses. Combined with formal analysis, this suggests Parker operates with chorus-level consistency but within-chorus development toward formal high points (bridges, turnarounds).

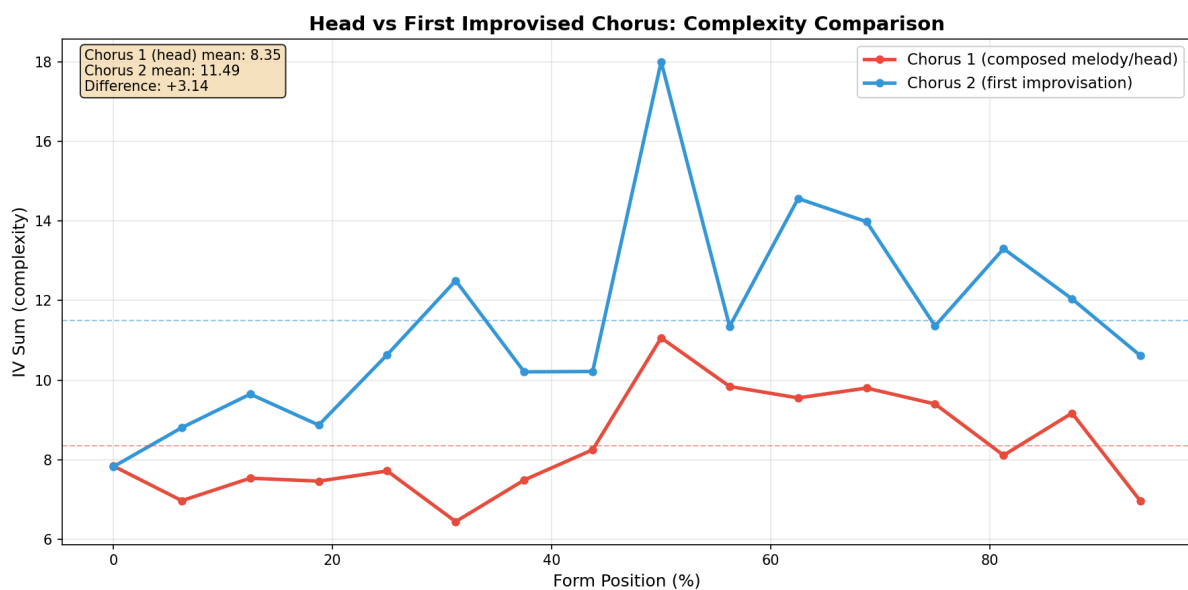


Figure 10: chorus 1 (head) vs chorus 2 complexity

## 4.9. Improvisational strategies

Autocorrelation analysis combined with volatility clustering reveals distinct temporal strategies:

- **Strategy 1: Balanced/Moderate (25 tunes)**
  - Example: “Barbados”
  - Moderate complexity (mean: 9.36)
  - Moderate volatility (CV: 0.577)
  - Positive autocorrelation (LAG-1: +0.260)
  - Characteristics: Consistent harmonic density with moderate variation, positive phrase-to-phrase continuity. Most common strategy in the corpus.
- **Strategy 2: Contrasting (13 tunes)**
  - Example: “Si Si”
  - Moderate complexity (mean: 10.70)
  - Moderate volatility (CV: 0.517)
  - Negative autocorrelation (LAG-1: -0.106)

- ▶ Characteristics: Phrase  $t$  negatively predicts phrase  $t + 1$ : Parker alternates between contrasting harmonic materials rather than developing continuously. Predictable unpredictability.
- **Strategy 3: Exploratory (7 tunes)**
  - ▶ Example: “Cosmic Rays”
  - ▶ High complexity (mean: 12.88)
  - ▶ High volatility (CV: 0.703)
  - ▶ Low autocorrelation (LAG-1: +0.192)
  - ▶ Characteristics: Sustained high harmonic density with dramatic variation. Low autocorrelation suggests episodic structure: phrases organized by chorus-level development arcs rather than local phrase-to-phrase relationships. Composed improvisation.
- **Outlier Tune: Volatile/Continuous (1 tune)**
  - ▶ Example: “My Little Suede Shoes”
  - ▶ Low complexity (mean: 6.58, lowest in corpus)
  - ▶ Extreme volatility (CV: 0.986, highest in corpus)
  - ▶ High autocorrelation (LAG-1: +0.529, highest in corpus)
  - ▶ Characteristics: alternates between extremely simple and moderately complex passages with high predictability. “Trap door” structure where volatility is self-perpetuating: complexity begets more complexity through phrase-level continuity, but overall complexity remains low through strategic simplicity.

These strategies are not conscious performer choices but emergent properties of tune, specific temporal organization. Parker conceives improvisational approaches holistically per composition.

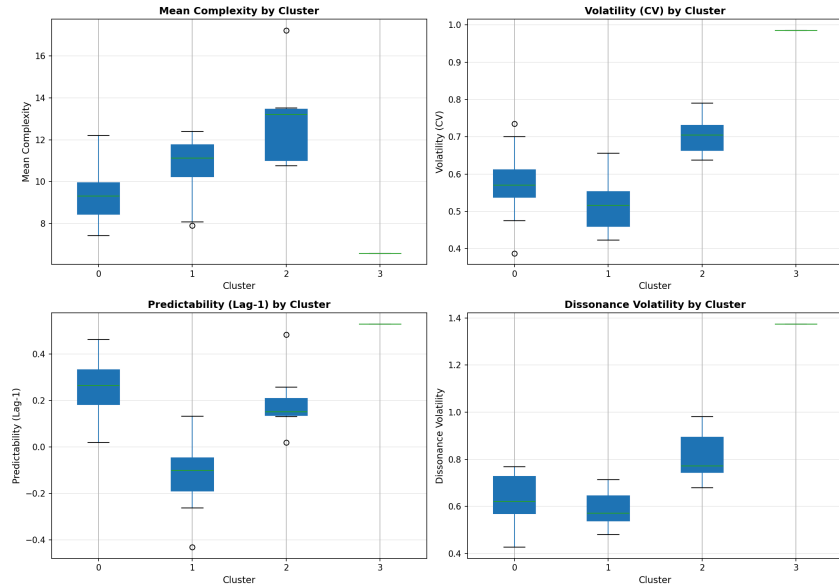


Figure 11: cluster characteristics showing separation across four features. Cluster 0 (“Balanced”,  $n=25$ ): moderate complexity with positive autocorrelation. Cluster 1 (“Contrasting”,  $n=13$ ): moderate complexity with negative autocorrelation. Cluster 2 (“Exploratory”,  $n=7$ ): highest complexity and volatility with near-zero autocorrelation. Cluster 3 (the outlier,  $n=1$ ): paradoxically combines lowest complexity with extreme volatility and high positive autocorrelation.

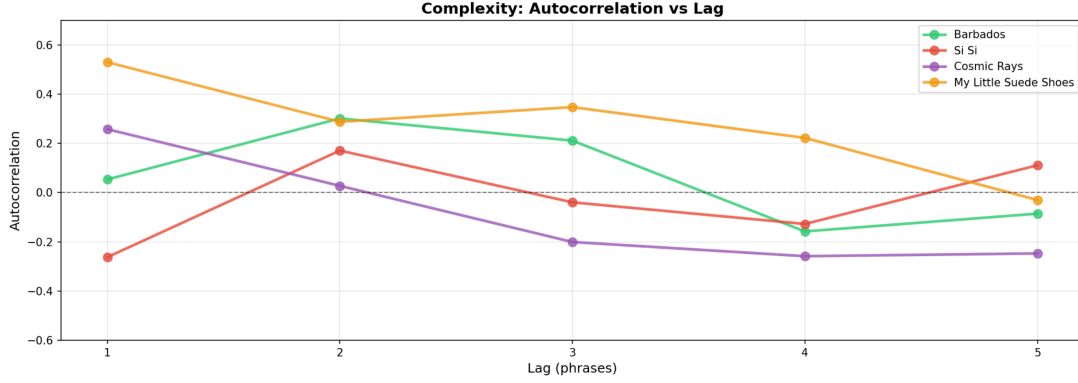


Figure 12: autocorrelation profiles demonstrating four improvisational strategies. “My Little Suede Shoes” (orange) maintains highest autocorrelation across all lags, suggesting trap door structure with predictable mode persistence. Si Si (red) begins negative, revealing contrasting phrase organization. Cosmic Rays (purple) starts low-positive but decays negative, consistent with episodic exploratory structure. Barbados (green) shows moderate positive autocorrelation characteristic of balanced strategy.

#### 4.10. Gravity

We introduced *gravity* to describe directional causal relationships between harmonic dimensions in improvisation, measured via Granger causality testing. For each tune with  $\geq 10$  phrases, we tested whether past values of one variable improve prediction of another variable beyond autoregressive history.

Granger causality testing reveals that phrase-level reactive processes account for a minority of temporal organization. In 17.4% of tunes (8/46), past values of dissonance significantly predict future complexity ( $p < 0.05$ ). The reverse direction shows weaker effects: 8.7% of tunes (4/46) show significant Complexity $\rightarrow$ Dissonance causality. Critically, 80.4% of tunes (37/46) show no significant causality in either direction, suggesting that most temporal organization operates independently of local phrase-level reactive patterns.

Among the 6 tunes exhibiting unidirectional causality, 5 show Dissonance $\rightarrow$ Complexity while 1 shows Complexity $\rightarrow$ Dissonance. This 5:1 ratio is suggestive of preferential directionality but does not reach statistical significance (binomial test: 5 of 6 unidirectional cases favor D $\rightarrow$ C,  $p = 0.109$ ,  $n = 6$ ). The mean F-statistic for Dissonance $\rightarrow$ Complexity (2.12) exceeds that for Complexity $\rightarrow$ Dissonance (1.36), supporting modest directional asymmetry, though the overlap in distributions suggests this is not a universal pattern.

Three tunes exhibit bidirectional causality, showing significant Granger effects in both directions simultaneously. This pattern suggests complex feedback loops where dissonance and complexity mutually influence each other across phrase boundaries, distinct from the unidirectional patterns observed in other performances.

**Strategy-specific patterns** reveal critical distinctions in temporal organization. Cluster 2 (“Exploratory”, 7 tunes) shows markedly reduced causality compared to other strategies: zero instances of Complexity $\rightarrow$ Dissonance causality (0/7 vs 8.0-15.4% in other clusters) and only one instance of Dissonance $\rightarrow$ Complexity causality (1/7, 14.3%). “Exploratory” performances also show zero phrase-length effects in either direction (0/7), contrasting with Cluster 1’s

23.1% (3/13). This pattern is consistent with composed improvisation: temporal organization determined at chorus level through predetermined developmental arcs rather than reactive phrase-level processes.

Cluster 0 (“Balanced”, 25 tunes) exhibits moderate Dissonance→Complexity causality (5/25, 20.0%), suggesting these performances blend reactive navigation with architectural planning. The phrase-level causality operates selectively rather than universally, suggesting real-time decision-making constrained by chorus-level structural intentions.

The outlier “My Little Suede Shoes” (Cluster 3) exhibits a distinctive pattern: zero Dissonance↔Complexity causality in either direction but perfect phrase-length causality (1/1, 100% in both directions). This tune’s “trap door” structure operates through phrase-length modulation rather than harmonic tension: volatility perpetuates through strategic alternation between brief and extended phrases rather than through dissonance-complexity feedback loops. “My Little Suede Shoes” exemplifies Parker’s engagement with Afro-Cuban rhythms and simplified harmonic forms, contrasting with his bebop-oriented tunes. The extreme volatility (CV = 0.99) despite low mean complexity (6.58) suggests strategic alternation appropriate to the tune’s rhythmic and formal character.

This finding reframes improvisation not as uniformly spontaneous but as strategically variable: Parker selects organizational modes, reactive navigation vs. composed architecture vs. phrase-length modulation, appropriate to each tune’s character and formal constraints. The “Exploratory” strategy’s near-absence of phrase-level causality (1/28 total significant effects across 7 tunes) suggests that predetermined chorus-level planning can operate independently of local reactive processes.

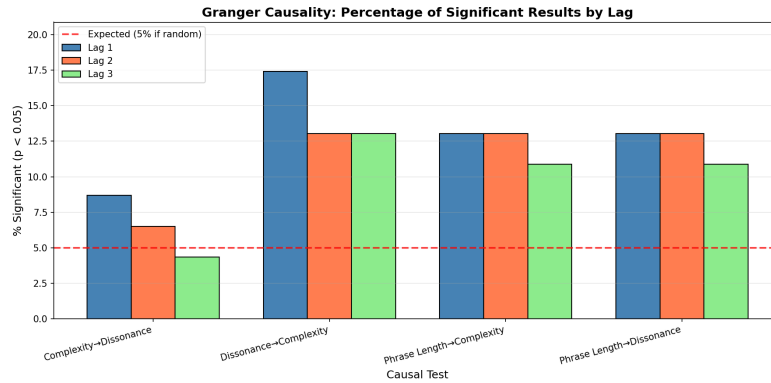


Figure 13: Granger causality significance rates across 46 tunes showing that phrase-level causality is a minority pattern (19.6% overall). Among tunes exhibiting causality, Dissonance→Complexity (17.4%) exceeds Complexity→Dissonance (8.7%), though asymmetry is suggestive rather than definitive ( $p = 0.109$ ).

### 4.11. Robustness

We introduced robustness to measure triadic content in interval vectors: how many complete triads (major, minor, diminished, augmented) are embedded in each pitch-class set. This metric tests whether frequently used interval vectors are triadically rich (offering stable harmonic foundations) or triadically sparse (offering navigational flexibility).

Results reveal a significant negative correlation between frequency and triadic content ( $r = -0.350$ ,  $p < 0.001$ ,  $n = 165$ ). Parker’s most frequently used interval vectors contain substantially fewer triads than less frequent IVs.

The top 20 most frequent interval vectors show mean triadic content of 0.6 triads per IV, compared to corpus mean of 2.6 triads per IV ( $t = -3.66$ ,  $p < 0.001$ , Cohen’s  $d = -0.87$ , 95% CI:  $[-2.55, -1.50]$ ). Frequency-weighted mean triadic content (1.09) falls substantially below the unweighted corpus mean (2.33), confirming Parker systematically favors triadically sparse structures.

Analysis across frequency quartiles (Table 3) reveals a monotonic pattern: rare IVs (Q1) contain mean 3.7 triads, while the most common IVs (Q4) contain only 0.9 triads (ANOVA:  $F = 12.13$ ,  $p < 0.001$ ). This gradient demonstrates that triadic sparsity correlates systematically with Parker’s usage preferences rather than occurring by chance.

| Frequency Quartile | Mean Triads | SD   | N  |
|--------------------|-------------|------|----|
| Q1 (Rare)          | 3.74        | 2.94 | 45 |
| Q2                 | 2.53        | 2.26 | 38 |
| Q3                 | 1.98        | 1.91 | 41 |
| Q4 (Most Common)   | 0.94        | 1.23 | 41 |

*One-way ANOVA:  $F(3, 161) = 12.13$ ,  $p < 0.001$*

Table 3: triadic content decreases monotonically with usage frequency. Rare interval vectors (Q1) contain more than twice the triadic content of frequently used IVs (Q4).

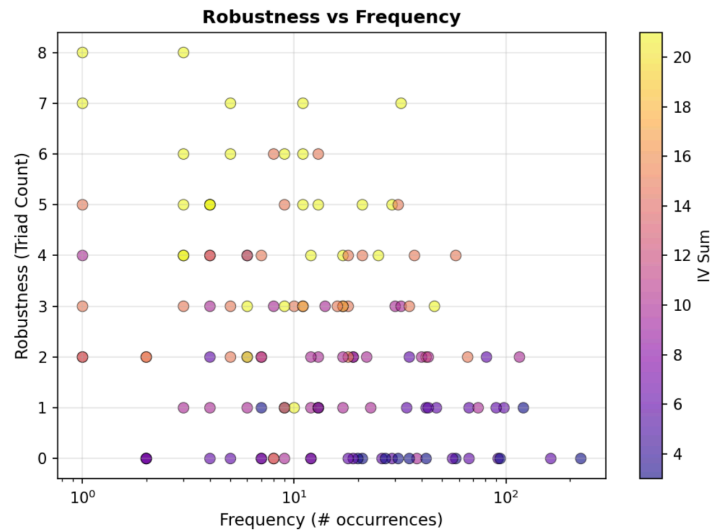


Figure 14: robustness (triadic content) as a function of usage frequency.

## 5. Discussion

### 5.1. Temporal evolution in bebop

Our findings suggest an alternative to the traditional assumption that improvisers start simply and progressively introduce more complex materials during performance, building toward climax through vocabulary expansion.

In Parker’s practice, vocabulary is largely complete before improvisation begins: 81.1% of unique interval vectors appear in the first improvised chorus. Temporal evolution is not accumulation of materials but exploration of pathways: new transitions between existing harmonic nodes. When complexity increases during performance, it’s not because new materials are introduced but because Parker navigates more rapidly or tortuously through existing materials.

This pattern aligns with cognitive models of expert improvisation emphasizing pre-learned chunks and schemas (Norgaard 2014; Pressing 1988). Norgaard’s research on improviser cognition demonstrates that expert jazz musicians rely on extensive pattern vocabularies stored in long-term memory, deployed through rapid retrieval and recombination rather than real-time generation of novel materials. Our finding that Parker’s vocabulary front-loading (81.1% in first chorus) supports this chunking model: improvisational fluency emerges from navigational facility through pre-established vocabulary rather than progressive invention during performance.

This reframing has pedagogical implications. Rather than drilling increasingly exotic scales assuming students will progressively deploy them during solos, pedagogy might benefit from emphasizing navigational fluency: how to move flexibly through materials you already know. The challenge is not “what to play” but “how to move.”

### 5.2. Temporal organization is tune-specific

The three improvisational strategies reveal that temporal organization is tune-specific, not performer-general. Parker doesn’t apply a single approach across all contexts; rather, he conceives distinct temporal architectures for different compositions.

This challenges assumptions about “personal style” as consistent approaches applied uniformly. While Parker’s intervallic vocabulary is recognizable across tunes (certain interval vectors appear frequently regardless of composition), his temporal organization varies dramatically. “Cosmic Rays” uses episodic structure, “Bluebird” uses slow-burn climax, “My Little Suede Shoes” uses trap door contrasts, “Si Si” maintains consistency.

This suggests improvisation operates at multiple scales with different organizing principles:

- Material level (which interval vectors): relatively consistent across tunes
- Temporal level (how material unfolds): tune-specific strategic architectures
- Phrase level (surface articulation): consistently short across all tunes

The interaction between these scales produces the rich temporal complexity we observe in expert improvisation.

### 5.3. Reactive navigation and temporal heterogeneity

The Granger causality findings reveal fundamental heterogeneity in Parker’s temporal organization strategies. The majority of performances show no significant phrase-level causality between dissonance and complexity, suggesting that temporal organization in these tunes operates according to principles other than local reactive processes. This finding challenges assumptions

about improvisation as uniformly “reactive” and instead suggests multiple organizational modes coexisting within a single performer’s practice.

### 5.3.1. Strategy-specific causality patterns

Strategy-specific patterns provide direct evidence for distinct organizational logics. Cluster 2 (“Exploratory”, 7 tunes) exhibits markedly reduced causality: only 1 significant result across 28 possible tests (3.6%), compared to 11/100 (11.0%) for Cluster 0 (“Balanced”) and 10/52 (19.2%) for Cluster 1 (“Contrasting”). This near-absence of phrase-level causality in “Exploratory” performances indicates what we term *composed improvisation*: temporal organization determined at chorus level through predetermined developmental arcs operating independently of local reactive processes.

“Balanced” performances (Cluster 0, 25 tunes) exhibit moderate Dissonance→Complexity causality (5/25, 20.0%), suggesting these performances blend reactive navigation with architectural planning. Phrase-level causality operates selectively rather than universally, revealing phrase-level navigation guided by larger-scale organizational goals.

“Contrasting” performances (Cluster 1, 13 tunes) show elevated phrase-length effects (3/13, 23.1% for both Phrase Length→Complexity and Phrase Length→Dissonance), suggesting temporal organization operates partially through phrase duration modulation rather than purely through harmonic tension. This represents a distinct organizational mode where volatility emerges from strategic phrase-length variation rather than harmonic feedback loops.

The outlier “My Little Suede Shoes” exemplifies yet another organizational logic: zero Dissonance↔Complexity causality but maximal phrase-length causality (1/1, 100%). This tune’s “trap door” structure operates through phrase-length modulation rather than harmonic navigation, demonstrating that organizational modes can be tune-specific rather than strategy-general.

### 5.3.2. Reactive navigation in the minority

Among the minority of tunes exhibiting phrase-level causality (19.6%), patterns are consistent with—though do not definitively prove—reactive navigation. When *dissonance* at phrase  $t$  predicts *complexity* at phrase  $t+1$ , this could reflect several processes: (1) reactive navigation, where Parker perceives *dissonance* as requiring resolution through *complexity*; (2) harmonic constraint, where the chord progression creates conditions where dissonant choices constrain subsequent options; (3) compositional planning, where Parker pre-plans both dissonant passages and resolutions at chorus level, creating statistical association without phrase-level causation; or (4) transcription artifacts from segmentation methodology.

The 2:1 ratio favoring Dissonance→Complexity over Complexity→Dissonance (17.4% vs 8.7%) suggests preferential directionality, though the binomial test ( $p = 0.109$ ) indicates this asymmetry does not reach statistical significance. Combined with higher mean F-statistics for Dissonance→Complexity (2.12 vs 1.36), the pattern is consistent with Parker responding to dissonant conditions through complexity more frequently than creating dissonance via complex choices, though the evidence is suggestive rather than conclusive.

### 5.3.3. Cognitive frameworks for temporal organization

This heterogeneity connects to existing cognitive models of improvisation. Pressing’s (Pressing 1988) cognitive constraints model posits that improvisers operate under real-time processing limitations requiring strategic management of cognitive load. Pressing identifies “event clusters”—points of high density requiring intensive processing—as natural sites for cognitive

bottlenecks. Our Granger causality findings suggest that dissonant harmonic conditions function as such bottlenecks: encountering dissonance triggers subsequent complexity as Parker navigates out of perceptually unstable regions. The directional asymmetry (dissonance predicting complexity more than the reverse) is consistent with Pressing’s model where harmonic tension creates processing demands that constrain subsequent choices.

However, the predominance of performances showing no causality indicates that reactive processing operates selectively rather than universally. Most performances proceed according to pre-planned architectures immune to phrase-level reactive constraints, suggesting Parker’s expertise enables flexible switching between reactive and compositional modes.

Johnson-Laird (Johnson-Laird 2002) distinguishes between “algorithmic” improvisation—rule-based generation proceeding step-by-step through local decision-making—and “inspirational” improvisation—spontaneous expression of pre-conceived musical ideas. Our findings suggest Parker employs both modes strategically: the performances showing no phrase-level causality align with Johnson-Laird’s inspirational mode, where chorus-level architectures unfold independently of local reactive processes. The 19.6% exhibiting Granger causality represent algorithmic mode, where harmonic conditions at moment  $t$  constrain choices at moment  $t+1$  through rule-based navigation.

This framework helps explain the strategy-specific patterns: “Exploratory” performances (3.6% causality) operate primarily through inspirational mode, executing predetermined developmental arcs; Balanced performances (11.0% causality) blend both modes; “Contrasting” performances emphasize phrase-length variation as a structural algorithm distinct from harmonic reactive processes.

#### 5.3.4. Synthesis: multiple organizational logics

Rather than a single organizational mode (reactive vs. composed, spontaneous vs. planned), expert improvisers like Parker operate across a continuum, selecting organizational strategies appropriate to each performance. The term *gravity* captures this phenomenon: in some performances, harmonic tension exerts directional force on subsequent choices (reactive/algorithmic mode); in others, temporal organization proceeds independently of local conditions (composed/inspirational mode); in still others, phrase-length variation drives temporal structure independently of harmonic content (structural algorithm).

The strategy-specific causality patterns - “Exploratory” showing near-zero effects (3.6%), “Balanced” showing moderate effects (11.0%), “Contrasting” emphasizing phrase-length (23.1%) — demonstrate that organizational modes correlate with broader improvisational strategies identifiable through autocorrelation and volatility analysis. The minority status of phrase-level causality overall (19.6%) suggests that chorus-level architectural planning predominates over moment-to-moment reactive navigation in Parker’s mature improvisational practice. Mastery emerges not from consistent application of a single organizational principle but from flexible deployment of multiple temporal logics appropriate to compositional context and improvisational intent.

#### 5.4. Triadic sparsity

The negative correlation between frequency and triadic content ( $r = -0.350$ ,  $p < 0.001$ ) challenges fundamental assumptions about bebop harmony. Traditional harmony assumes triads provide stable foundations above which extensions are added. In jazz harmony, triads also make great “upper structure” to super-impose over harmonic material. Parker’s practice reveals

a different organizing principle: his most frequently deployed interval vectors are triadically sparse, not triadically rich.

The magnitude of this preference is substantial: Parker’s most common IVs (top 20) contain less than one quarter the triadic content of the corpus average (0.6 vs 2.6 triads, Cohen’s  $d = -0.87$ ). This represents a large effect size, suggesting systematic selection rather than random variation. When weighted by actual usage frequency, Parker’s effective triadic content (1.09) falls 53% below the available corpus mean (2.33).

This pattern suggests bebop harmony is organized not around triadic stability but around navigational flexibility. Triadically sparse interval vectors offer more degrees of freedom for continuation: they don’t commit to specific triadic implications, enabling pivots to multiple harmonic destinations. This explains why Parker’s frequently used interval vectors serve as hubs in temporal networks: they’re selected for connectivity rather than harmonic stability. An IV like (111000), containing only one triad despite six interval classes, functions effectively precisely because it avoids triadic commitment.

Parker’s practice instead suggests mastery emerges from navigating between harmonic regions through triadically ambiguous pivot structures rather than from accumulating exotic triadic extensions. The correlation between triadic sparsity and usage frequency ( $r = -0.350$ ) indicates this is a deliberate preference: Parker systematically selects pivot structures over stable foundations.

## 5.5. Phrase-level vs. chorus-level hierarchy

The finding that phrases are extremely short (median 1 segment) while meaningful patterns emerge at 5-phrase window level reveals hierarchical temporal organization:

- Surface level: Rapid articulation (1-2 segments per phrase)
- Phrase-group level: Meaningful harmonic motion (5-phrase windows half-chorus)
- Chorus level: Strategic architecture (solo development arcs, overall complexity)

This hierarchy parallels linguistic organization: phonemes  $\rightarrow$  words  $\rightarrow$  phrases  $\rightarrow$  sentences. Each level has distinct organizing principles:

- Surface: Breath/articulation determines phrase boundaries
- Phrase-group: Local harmonic motion and reactive navigation
- Chorus: Strategic architecture and composed plans

The three improvisational strategies operate at chorus level, not phrase level. “My Little Suede Shoes” is an outliner and alternates between simple and complex **phrase groups** (Strategy 2: “Contrasting”), not individual phrases. This suggests Parker’s improvisational conception operates primarily at phrase-group and chorus levels, with surface articulation following from these higher-level plans.

## 6. Conclusion

This study demonstrates that time series analysis reveals temporal organization principles in jazz improvisation invisible to traditional analytical approaches. By tracking harmonic metrics (*complexity, dissonance, rate of change*) across temporal scales from segments through phrases to choruses, we uncover strategic architectures, causal relationships, and hierarchical organization that challenge pedagogical assumptions.

Key findings reframe our understanding of bebop improvisation:

1. **Temporal heterogeneity predominates over uniform reactive processes:** majority of performances show no phrase-level causality between dissonance and complexity, suggesting that chorus-level architectural planning, what we term *composed improvisation*, constitutes Parker’s primary organizational mode rather than moment-to-moment reactive navigation.
2. **Three distinct improvisational strategies exhibit different temporal logics:** “Exploratory” performances (7 tunes) show near-zero phrase-level causality (3.6%), consistent with predetermined developmental arcs; “Balanced” performances (25 tunes) exhibit moderate reactive effects (11.0%); “Contrasting” performances (13 tunes) emphasize phrase-length modulation (23.1%). Temporal organization is tune-specific rather than performer-general, with Parker selecting organizational modes appropriate to compositional context.
3. **Vocabulary front-loading enables pathway exploration:** 81.1% of unique interval vectors appear in the first improvised chorus, with 16.5% added during the second chorus and minimal additions thereafter. Temporal evolution reflects exploration of new transitions between existing harmonic nodes rather than progressive vocabulary expansion, though the substantial second-chorus addition indicates refinement continues beyond initial deployment.
4. **Triadically sparse structures predominate in frequent usage:** significant negative correlation between frequency and triadic content ( $r = -0.350$ ,  $p < 0.001$ ) reveals that Parker’s most common interval vectors contain less than half the triadic content of corpus average (0.6 vs 2.6 triads, Cohen’s  $d = -0.87$ ). Mastery emerges from navigational flexibility through triadically ambiguous pivot structures rather than accumulation of triadically rich exotic materials.
5. **Hierarchical temporal organization operates across multiple scales:** surface articulation (relatively brief phrases averaging 2 segments) organizes into phrase groups (5-phrase windows revealing meaningful patterns) organized by chorus-level strategic architecture, with different organizing principles operating at each temporal scale.

We introduce novel analytical concepts: *gravity* (directional causal relationships measured via Granger testing) and *robustness* (triadic content as structural flexibility). These concepts enable quantitative analysis of improvisational practice and bridge computational analysis and music-theoretical interpretation, revealing how improvisers navigate harmonic space through multiple organizational logics.

## 6.1. Limitations

This study’s findings are subject to several methodological constraints. First, interval vectors abstract away voice-leading and temporal ordering information, capturing pitch-class content rather than melodic contour. While appropriate for examining harmonic resource selection, this abstraction cannot address questions of linear motion or registral organization.

Second, phrase detection relies on transcribed rest notation, which represents transcriber interpretation of recorded performance. Alternative segmentation methods (e.g., dynamic thresholding, machine learning boundary detection) may yield different phrase structures.

Third, Granger causality testing measures predictive asymmetry rather than causal intent. While significant results indicate that dissonance at phrase  $t$  improves prediction of complexity at phrase  $t + 1$  beyond autoregressive history, this reflects statistical association within performance time series rather than cognitive causation.

Fourth, dissonance weighting ( $ic1 \times 1.0$ ,  $ic6 \times 0.8$ ,  $ic2 \times 0.5$ ) reflects psychoacoustic principles but remains parameterized. While alternative weighting schemes would alter absolute dissonance values, strategy identification relies on *volatility* (coefficient of variation) rather than absolute levels, providing robustness against moderate weight adjustments. Future work should systematically test sensitivity to weighting schemes, though preliminary testing with alternative weights ( $ic6 = 0.6$  and  $1.0$ ) produced qualitatively similar strategy assignments.

Finally, this corpus represents Parker’s recorded output as transcribed in the Omnibook, not the complete space of his improvisational practice. Findings characterize this corpus of Parker’s recorded improvisations and should be tested against other artists and repertoires before generalization.

## 6.2. Future directions

This methodology enables several future research directions:

- **comparative analysis:** Extend to multiple artists, revealing whether findings are Parker-specific or constitute broader bebop principles
- **multilayer networks:** Integrate pitch, rhythm, and harmonic function layers, revealing interactions invisible in single-dimension analysis
- **predictive modeling:** Use temporal patterns to predict next harmonic choices, testing whether computational models can capture improvisational logic
- **pedagogical applications:** Develop practice strategies emphasizing navigational fluency and hub-centric motion rather than vocabulary accumulation
- **historical evolution:** Track how temporal organization strategies changed across bebop’s development from 1940s through 1950s

The framework developed here, particularly *gravity* and *robustness* concepts, three strategic categories, and multi-scale temporal analysis, provides tools for such comparative work. Time series analysis of jazz improvisation reveals hidden temporal structures that connect musical practice to broader principles of cognitive organization and creative behavior.

## 7. Data and code availability

The Charlie Parker Aligned Digital Omnibook dataset combines manual transcriptions (Déguer-  
nel, Vincent, Assayag) with automatic transcription methods and MIDI alignment (Riley,  
Dixon). MusicXML data provided by Inria and STMS Lab Ircam/CNRS/UPMC. Original  
copyrights held by Atlantic Music Corp. Data provided under Creative Commons Attribution-  
NonCommercial-ShareAlike license version 2.0.

All analysis code, processed datasets, and documentation are publicly available at  
[github.com/code91/parker-timeseries](https://github.com/code91/parker-timeseries) under MIT license. The repository includes the complete  
analytical pipeline, phrase detection algorithms, and visualization tools. All analysis performed  
using Python with pandas, NumPy, matplotlib, seaborn, NetworkX, and statsmodels libraries.

## 8. References

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